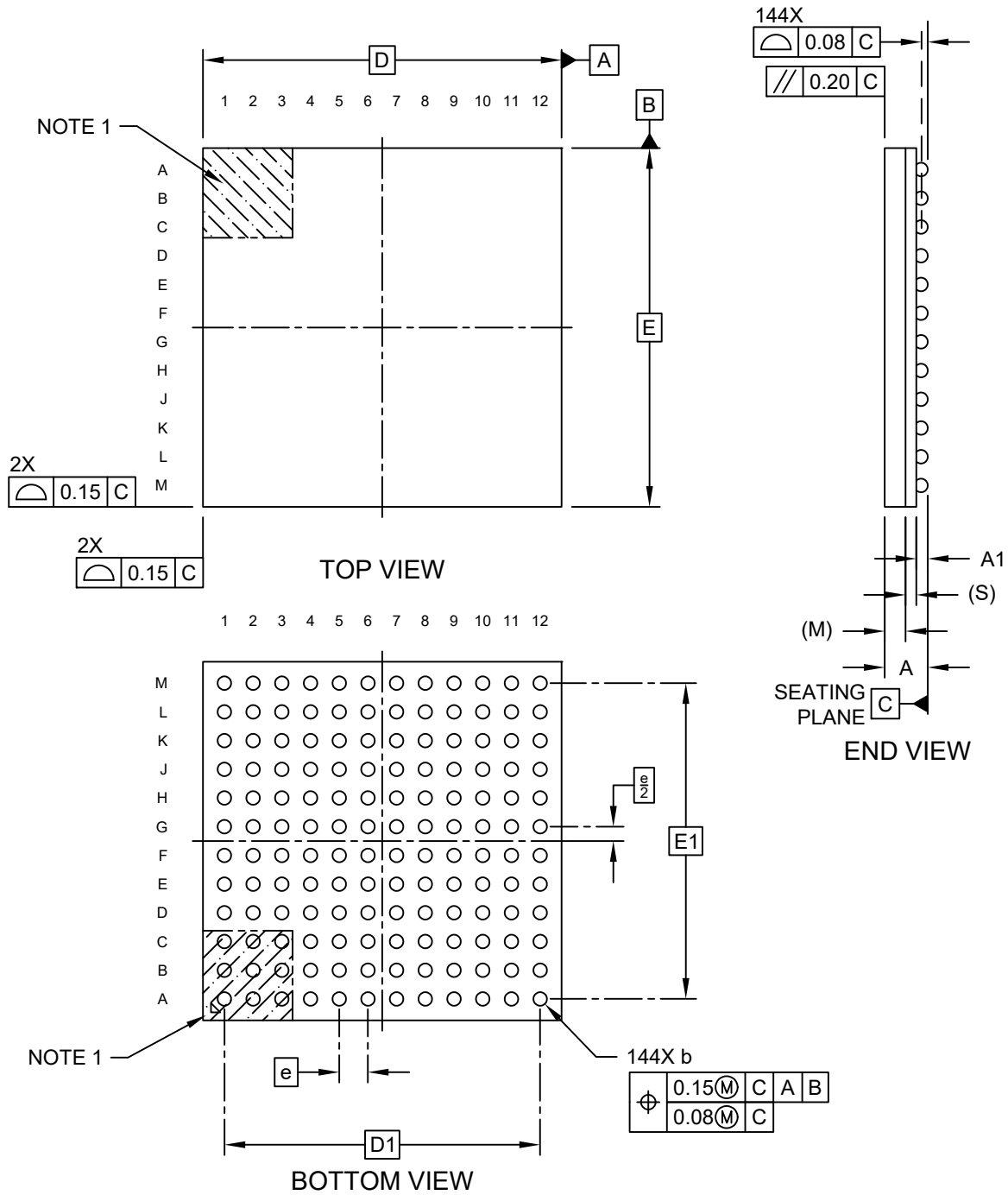


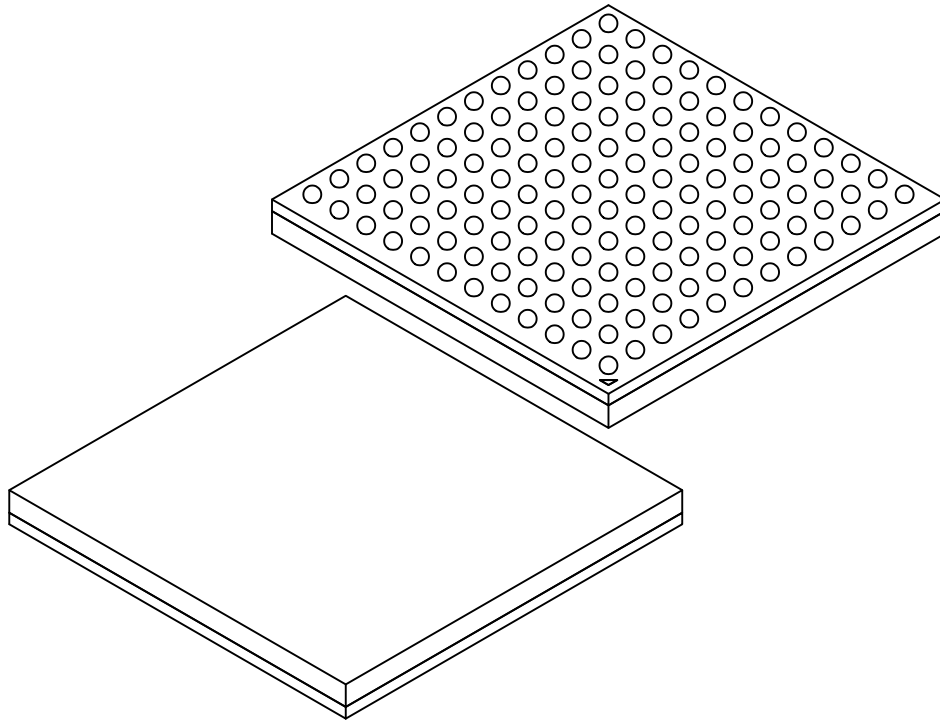
144-Ball Thin Fine-Pitch Ball Grid Array Package (HQB) - 10x10x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



144-Ball Thin Fine-Pitch Ball Grid Array Package (HQB) - 10x10x1.2 mm Body [TFBGA]

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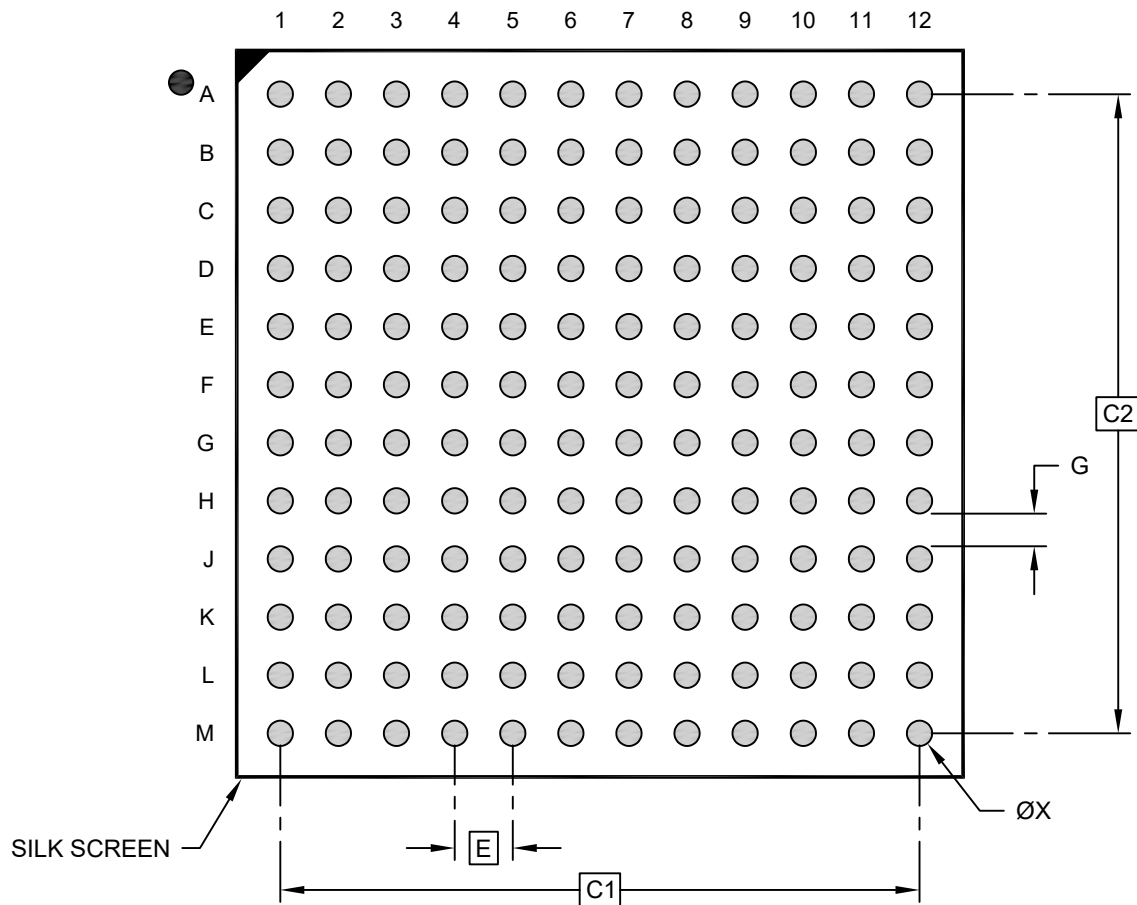
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.22	–	0.32
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.246 REF		
Overall Length	D	10.00 BSC		
Ball Array Length	D2	8.80 BSC		
Overall Width	E	10.00 BSC		
Ball Array Width	E2	8.80 BSC		
Ball Diameter	b	0.32	–	0.42

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Ball Thin Fine-Pitch Ball Grid Array Package (HQB) - 10x10x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	8.80 BSC		
Contact Pad Spacing	C2	8.80 BSC		
Contact Pad Diameter (X144)	X			0.35
Contact Pad to Contact Pad	G	0.45		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23565 Rev A